
NE  BOTIX

MP-400

Neobotix GmbH

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Operating Instructions

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CHAPTER 1

MP-400

↓ Download as PDF¹



The mobile robot MP-400 (and all varieties and models based on it) is an autonomous robot vehicle for a wide range of applications.

Its central differential drive with two big drive wheels provides high manoeuvrability on different surfaces and exact, efficient runs across longer distances. Measurement data from the laser scanner can be used for localisation, navigation and collision avoidance. This scanner also monitors different user-defined safety fields in front of the robot. As soon as an object is detected within the currently activated field, the robot is immediately set to emergency stop.

These features enable the robot to exactly determine its current position within the working area, to plan paths to arbitrary positions and to safely avoid collisions with passers-by or other dynamic obstacles.

Additional components and systems can be mounted onto and into the robot. They can be powered by the robot's internal power supply and can be controlled by the on-board computer.

¹ <https://neobotix-docs.de/hardware/en/MP-400.pdf>

1.1 Product Information

1.1.1 Intended Use

The mobile robot has been designed for daily operation in factory workshops and test halls. It can be used for transportation of materials, parts and devices. Furthermore the mobile robot can easily be used in research projects as mobile carrier of sensor equipment, robot arms and other special devices.

Depending on the intended application the mobile robot can be used on its own, in combination with other robot vehicles and in combination with stationary systems. Furthermore, application specific extensions can be integrated into the basic platform. These might be a customised cargo area, a robot arm or special sensors.

The mobile robot has been designed with focus on indoor operation. It is not recommended to operate the mobile robot outdoors, especially during rain, fog or snow.

1.1.2 Improper Use

Danger:

- The mobile robot must not be used for transportation of passengers in any way. No person must ride on the robot itself, nor must the robot be used to move any other vehicle or hanger with passengers aboard.
- The mobile robot must not be operated in any publicly accessible area without assessment of risks. If the robot is to be modified either permanently (e.g. by mounting additional components) or temporarily (e.g. by loading cargo) its safe operation must be assessed and approved.
- Without the safety approval described above, the mobile robot must at no times be used without supervision of a qualified operator if there are guests, passers-by or other people unfamiliar with the robot within the working area.
- The robot must never be operated in areas where there are staircases leading down, elevated platforms or other possibilities for falling or dropping down. This may cause serious injuries or death!

1.1.3 Working Area

The working area of the mobile robot must be protected against rain and have a sufficiently firm and clean floor. The floor must be even and horizontal.

Tip: Slopes, edges, steps and unevenness can result in problems when the robot is moving or with the localisation. Under some circumstances the robot might not be able to continuously determine its exact position and therefore might plan unnecessary or problematic paths or might even be unable to reach its destination.

Under certain conditions, rain or heavy splash water might enter the robot and cause damages. Flying sparks, heavy dust and similar dirt may affect or damage the sensors and thus might make the operation of the mobile robot inefficient or unsafe. Dirt and liquids on the floor may lead to slippage and problems of localisation and navigation.

The robot does not feature any sensor for detecting staircases, holes or other areas where it may drop down. In case of loss of localisation or faulty programming the robot might fall down such places.

Warning: Always check for and secure all places where the robot might drop down before bringing it into operation.

1.1.4 Qualified Personnel

This product must only be modified, commissioned, operated and serviced by qualified personnel.

The requirements on qualified personnel can be found at [Qualified Personnel](#) (page 31).

1.1.5 Safety Instructions

The general safety instructions for Neobotix robots can be found at [Safety Instructions](#) (page 24).

1.2 Operating Elements

The figures below show the sides of the MP-400 and the most important operating elements.

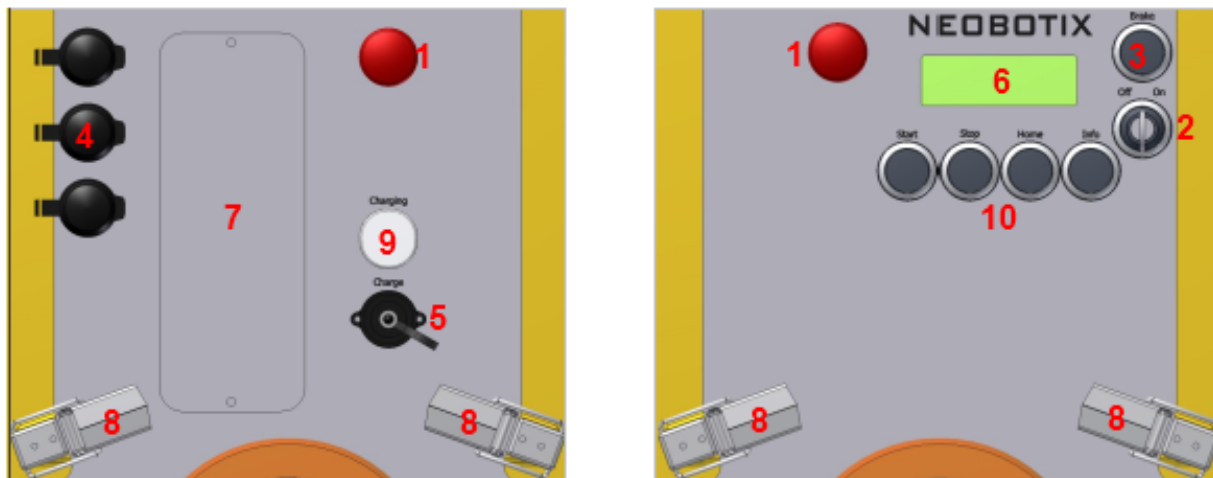


Fig. 1: Operating Elements

1	Emergency stop buttons
2	Key switch
3	Brake release button
4	USB / Ethernet sockets
5	Charging socket
6	LC-Display
7	Maintenance access (optional)
8	Cover fasteners
9	LED “charging”
10	Keypad

1.2.1 Emergency Stop Buttons

When one of these buttons is pressed the robot is immediately set to emergency stop. All drives are disconnected from power supply and the fail-safe brakes are engaged. This state can be reset by unlocking the emergency stop buttons and turning the key switch clockwise for a few seconds.

1.2.2 Key Switch

See *Key Switch* (page 13).

1.2.3 Brake Release Button

Pressing this button will open the motors' brakes thus allowing the robot to be moved manually even while it is turned off.

Tip: If the control software is running and the emergency stop buttons are not pressed, the motor amplifiers will still stabilise the robot's position when pressing the brake release button. In this case at least one of the emergency stop buttons must be pressed before the robot can be moved manually.

1.2.4 Charging Socket and Indicator LED

The battery charger is integrated into the robot. To charge the robot plug in the power cable and fasten it properly before connecting it to a normal AC power outlet. The white charging indicator LED will light up until charging is finished. The robot can remain connected to the power outlet even with the batteries fully charged.

1.2.5 LC-Display

This display shows the most important status information. A detailed description of the LCD can be found in *LC Display* (page 14).

1.2.6 Access to the On-Board Computer

The maintenance hatch provides access to the computer interfaces. A USB socket behind one of the black caps can be used to connect a wired joystick for temporary manual control in case there is no receiver for the wireless joystick plugged in already. An Ethernet socket is also available in case the WiFi connection is temporarily unavailable.

1.2.7 Keypad

The keypad can be used for simple input and predefined commands. By default it is without function and has to be interpreted by the application control software.

Pressing the `Info` button shows technical details of the robot on the LCD as described at *LC Display* (page 14).

1.3 Mechanical Properties

1.3.1 Dimensions

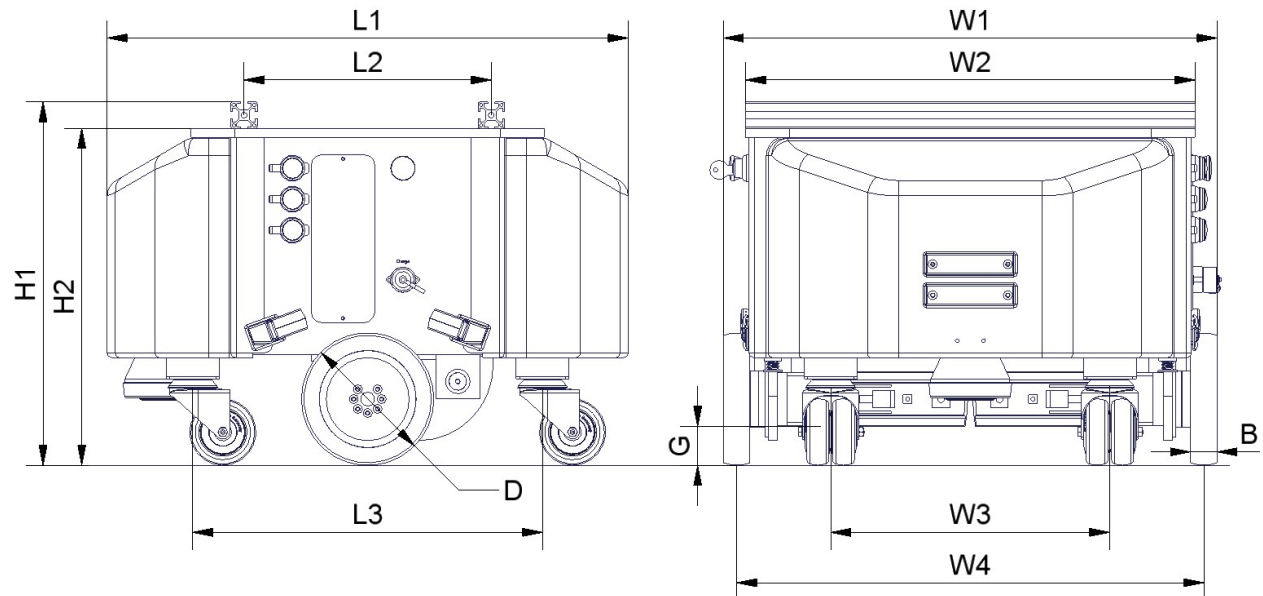


Fig. 2: Dimensions of the MP-400

Description	Symbol	Value (mm)
Diameter of drive wheels	D	150
Width of drive wheels	B	30
Ground clearance	G	44
Height of mounting profiles	H1	411
Height of top plate	H2	381
Overall length of platform	L1	590
Centre distance of mounting profiles	L2	280
Wheel base of the castor wheels	L3	396
Maximum width	W1	559
Width of mounting profiles and top plate	W2	509
Track width of the castor wheels	W3	316
Track width of the drive wheels	W4	529

1.3.2 Absolute Maximum Ratings

Warning: Exceeding these ratings might cause malfunctions or damage the robot!

Description	Units	Value
Payload (on top plate, standard version)	kg	100
Payload (on top plate, high payload version)	kg	150
Maximum speed	m/s	1.5
Maximum bumpiness to pass over ($\leq 0.25\text{m/s}$)	mm	10
Maximum bumpiness to pass over (full speed)	mm	5
Maximum acceleration	m/s^2	2.4
Storage temperature	$^{\circ}\text{C}$	-20 .. +60
Operating temperature (environmental temperature)	$^{\circ}\text{C}$	+0 .. +25

1.3.3 Positions of Sensors

All distances are in millimetres, measured relative to the platform's coordinate system. All angles are in degrees, measured counter-clockwise from the driving direction.

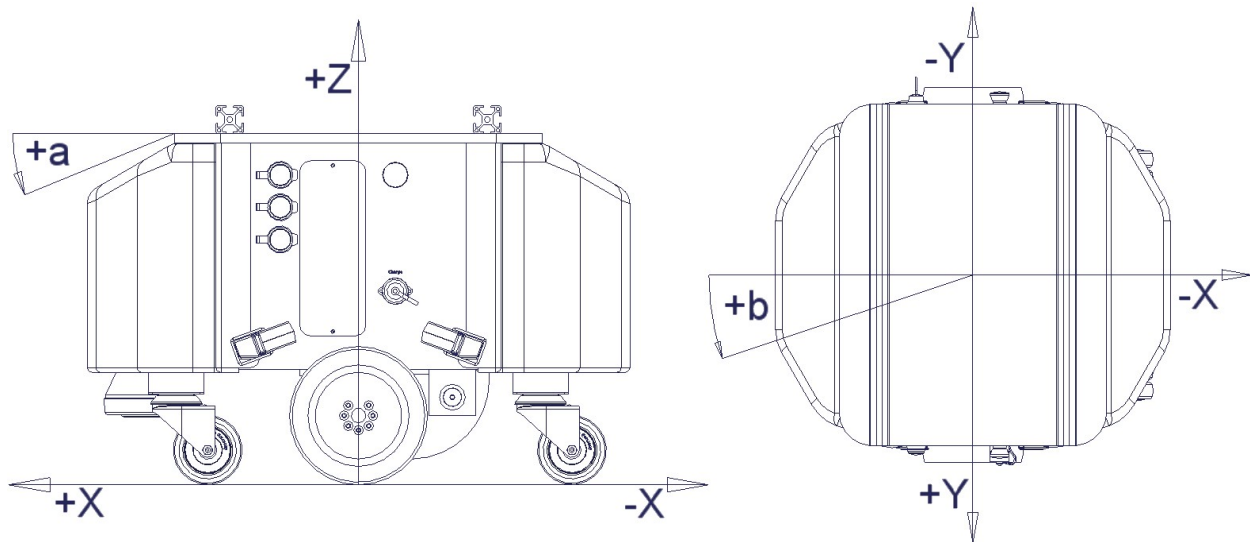


Fig. 3: Coordinate system of MP-400

Sensor	Symbol	X-pos.	Y-pos.	Z-pos.	a-angle	b-angle
Front laser scanner	LS1	230	0	110	0	0
Rear laser scanner	LS2	-354	0	110	0	180
Ultrasonic sensor 1	US1	-266	157	280	0	150
Ultrasonic sensor 2	US2	-298	0	280	0	180
Ultrasonic sensor 3	US3	-266	-157	280	0	210
Ultrasonic sensor 9	US9	266	157	280	0	30
Ultrasonic sensor 10	US10	298	0	280	0	0
Ultrasonic sensor 11	US11	266	-157	280	0	330

1.3.4 Electrical Properties and Miscellaneous Data

1.3.4.1 Properties of Internal Components

All data are taken from the respective data sheets.

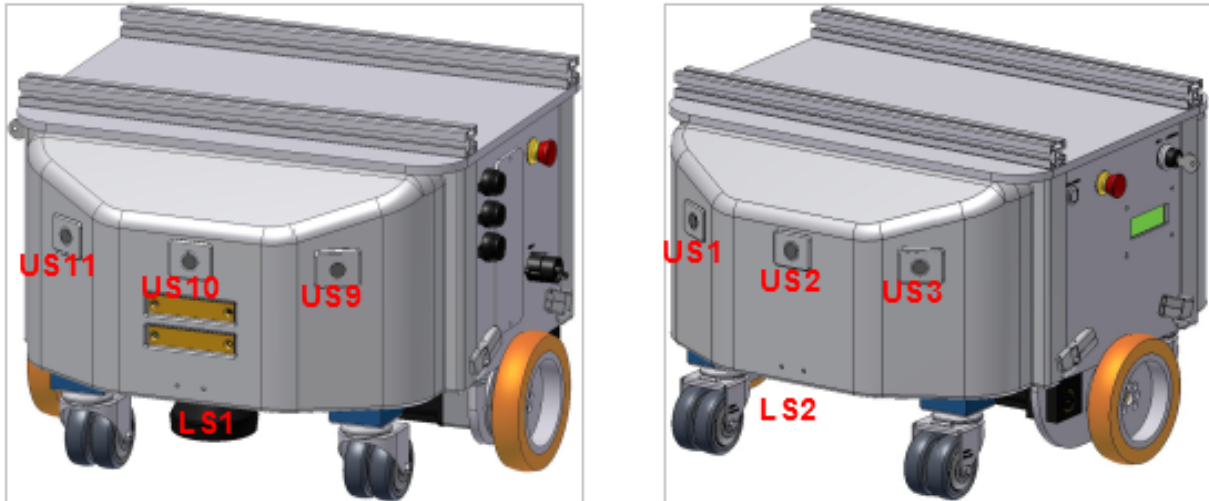


Fig. 4: Sensor positions at the MP-400

Description	Units	Value
Motor power	W	300
Rated motor speed	rpm	3000
Maximum motor speed	rpm	5000
Possible motor speed (at battery voltage)	rpm	1800
Rated motor torque	Nm	0.95
Maximum motor torque	Nm	2.85
Brake torque (static)	Nm	1.47
Encoder resolution	Increments/Revolution	10000
Gear ratio	1	8:1
Rated battery voltage	V	24
Maximum unregulated voltage	V	29
Battery capacity	Ah	50

1.3.4.2 Metering Capacities of Sensors

All data are taken from the respective datasheets.

Sensor	Resolution	Min.range	Max. range	Hor. angle	Vert. angle
Laser scanner	0.5 °	0 m	30 m	±135 °	0 °
Ultrasonic sensors	~±0.01 m	0.25 m	1.5 m	±60 °	±30 °

1.3.4.3 Other Properties

Description	Units	Value
Weight	kg	70
Expected working time	a	10

1.3.4.4 Connectors

See *Connectors* (page 22).

1.4 Transport

1.4.1 Packaging

The mobile robot MP-400 is packed in a rugged wooden box which can be reused for future transports. If the original box cannot be used anymore it is recommended to build a new box of similar design.

The base plate of the box has to be strong enough to carry the robot and to take asymmetric forces, e.g. when moving the box with a forklift. When packing the robot it has to be secured against unintended slipping by sufficient cushioning. In order to prevent dust and dirt from the cushioning material to enter the robot, the maintenance opening should be closed and all connectors should be fastened or covered prior to packing the robot.

Depending on the cushioning material it might be advisable to protect the brake release button against unintended activation. If this button is pressed during transport, the robot may move inside the box and the batteries might get discharged. Furthermore, the key of the key switch should be removed and transported separately to avoid damages to the key and the switch. Further information on the control elements can be found at *Operating Elements* (page 3).

1.4.2 Unpacking the Robot

If the robot is shipped in the original transport box only the lid should be removed. Then all small components and the cushioning material can be taken out.

Once the robot is free of all cushioning it can be lifted out of the transport box. This should be done by at least two people taking hold of the base plate of the platform. The best places for holding the robot are the areas beside the castor wheels.

Warning: Never try to lift the robot by holding it at the laser scanner or the control elements! None of these components has been designed for lifting forces and they may be damaged.

1.4.3 Short Distance Transport

For short distances the MP-400 can also be transported in the trunk of a car without the wooden box. It still must be secured against sliding and should only be moved very carefully while outdoors.

1.4.4 Long Distance Transport

As long as the robot is packed properly there are no special requirements when shipping the MP-400. Only in case the robot has been exposed to very low temperatures (below -10°C) it should be left to reach room temperature again before turning it on. This will prevent damages to the electrical and electronic components.

If the brake release button was covered and the key switch was removed there is no risk of the robot being turned on during the transport or the batteries being discharged.

Tip: To eliminate every risk when shipping the robot as airfreight it is recommended to disconnect the batteries prior to shipping. The batteries are leak proof and approved for airfreight.

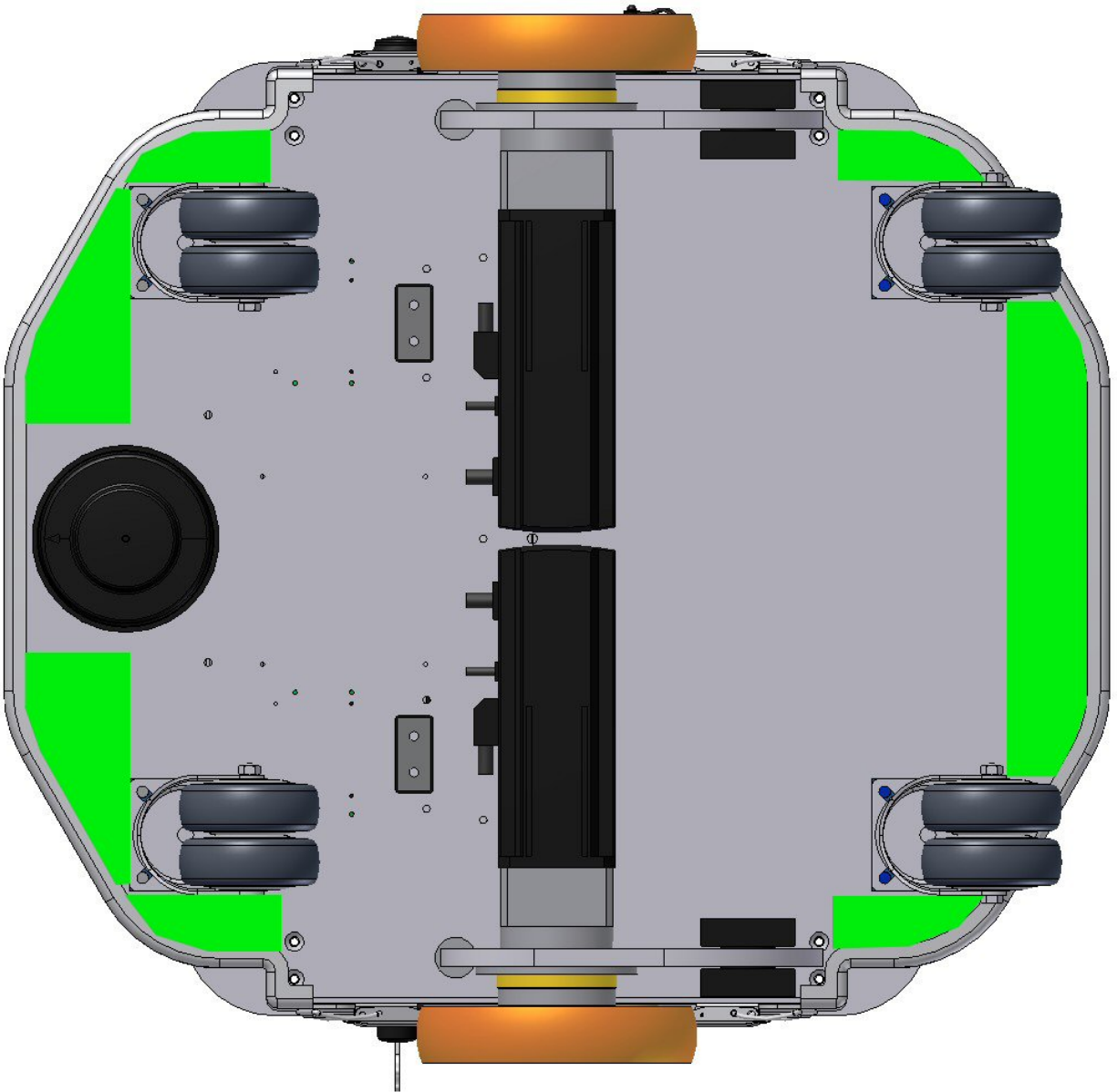


Fig. 5: Gripping areas of the MP-400

1.5 Maintenance

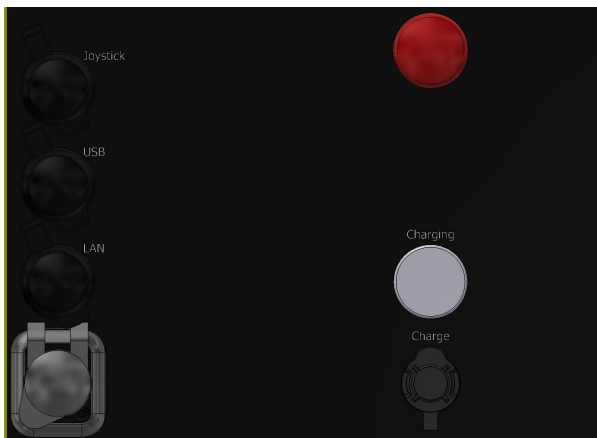
Information on hardware maintenance, cleaning and general precautions when working on the mobile robot can be found in [Maintenance](#) (page 30).

The robot's hardware is almost maintenance-free and the platform is not intended to be opened unless when used for research. Please contact Neobotix before disassembling the robot!

1.5.1 Software

Apart from research or maintenance it will not be necessary to connect mouse or keyboard because the robot can be controlled via wireless LAN and the graphical user interface. Even most software updates can be performed by remote access.

In order to access the platform's on-board computer directly, a USB and an Ethernet socket are available under the black caps at the left side of the robot.



1.5.2 Cleaning

To achieve a long life of the platform and guarantee a satisfying operation, the platform should be checked and cleaned regularly. Information on how to clean the robot and its sensors can be found in [Maintenance](#) (page 30).

1.5.3 Batteries

The platform's batteries provide enough capacity for about 10 h of continuous operation or covering up to 8 km.

Further information on the available battery type, as well as on battery maintenance and recycling, can be found [Batteries](#) (page 16).

In case you want to replace the batteries please follow these steps:

1. Switch off the robot.
2. Switch off and disconnect the battery charger. If the battery charger is installed inside the robot, disconnect the power cable from the robot.
3. Open the two quick release fasteners holding the cover at the rear side of the platform (the side without scanner and charging contacts) and open the cover.
4. Disconnect both battery connectors.

5. Remove the screws of the positioning bar behind the batteries.
 6. Lift the batteries out of the platform while taking care not to damage any cables.
 7. Move the new batteries into the platform while making sure that no cables get caught or damaged.
 8. Replace the positioning bar and fix it with the screws.
 9. Plug in the battery connectors.
 10. Close the cover and secure both fasteners.
- In case of any problems, please contact Neobotix.

1.5.4 Fuses

The mobile robot features the following fuses:

ID	Circuit	Location	Type	Current Rating	Characteristics
F1	24 V Terminals	RelayBoard	Micro 5x20	5 A	slow
F2	Motor Power	RelayBoard	Micro 5x20	10 A	slow
F3	Emergency Stop Buttons	RelayBoard	Micro 5x20	2 A	fast
F4	Brake Release Button	RelayBoard	Micro 5x20	2 A	fast

1.6 Taking out of Service

You can find information about taking out of service and about recycling [here](#) (page 32).

1.7 Legal Notes

The general legal notes can be found at [Legal Notes](#) (page 33).

1.7.1 EU Declaration of Conformity



This product fulfils all relevant directives of the European Union.

1.7.2 RoHS Information



As *non-road mobile machinery* our mobile robots are explicitly exempted from the scope of the RoHS directive 2011/65/EU. We have nevertheless followed the underlying principles of RoHS and tried to reduce the use of harmful and dangerous substances in our products as much as possible.

General Hardware Information

2.1 Key Switch

All Neobotix robots are equipped with a key switch to turn the robot on and off and to reset emergency stops. It can only be operated as long as the key is inserted.

2.1.1 Turning on

To start the robot, turn the key clockwise (towards position **II**) until the LCD lights up and then release again.

2.1.2 Emergency Reset

After an emergency stop was issued and the emergency buttons have been reset, turn the key clockwise (towards position **II**), hold for about one second and then release again. The message in the LCD display should change if the robot can be moved again.

2.1.3 Shut down

Turn the key counter-clockwise (towards position **I**) for five seconds to initiate the shutdown which will happen within 30 seconds. In this time, the platform software (if running) terminates and shuts down the operating system.

In case the on-board computer is shut down remotely (via the GUI or remote access), the robot will turn off automatically.

2.1.4 Switch off

To immediately turn off the robot without waiting for the on-board computer to shut down, turn the key switch counter-clockwise (towards position **I**) and hold it until the LCD light switches off.

Note: With this method, the robot is shut down by cutting the power supply. This may damage the on-board computer's hard- and software.

2.2 LC Display

2.2.1 Main View



2.2.1.1 Messages

The first line can be used to print messages sent from ROS, PlatformPilot or any other high level control software. The default message is “NEOBOTIX GmbH”.

2.2.1.2 Status

The second line indicates the robot's current status.

PLEASE WAIT The control software is not yet connected to the robot's main control board.

READY The control software is active and the robot is fully operational.

EMERGENCY STOP One of the emergency stop buttons has been pressed.

TURN KEY → Reset the safety system by turning the key switch clockwise for about one second.

SCANNER STOP An obstacle has been detected within the laser scanner's safety field. As soon as the obstacle has been removed, the safety system will be reset automatically. If a Flexi Soft safety controller is installed, all stops are indicated with this message.

CHARGING The robot is currently charging its batteries.

SHUTDOWN IN XXs The robot will turn itself off in a few seconds.

BATTERY VOLTAGE LOW The battery voltage is critically low and the robot will soon shut down to prevent a deep discharge. Further information can be found in the next section.

MOTOR ERROR At least one motor is reporting an error. This state is quit automatically as soon as all motors are operational again.

SAFETY RELAY FAIL One of the safety relays is damaged. Please contact Neobotix.

POWER RELAY FAIL The power relays are damaged. Please contact Neobotix.

EMSTOP BUTTON FAIL One of the emergency stop buttons does not operate properly. Please contact Neobotix.

CHARGE RELAY FAIL The charging relay is damaged. Please contact Neobotix. The charging contacts might still be connected to the batteries!

CHECKSUM ERROR The protocol versions of the control software and the RelayBoard do not match. Please check the wiring and the log output of your software.

2.2.1.3 Battery Charge Level

B:096% The current charge level of the batteries. Due to the batteries characteristics the actual battery voltage may be higher than the platform's rated main voltage.

Nominal voltage	Shut-down	Working range	Charging
24 V	≤ 22 V	23 V .. 25 V	25 V .. 29 V
48v	≤ 44 V	46 V .. 52 V	52 V .. 59 V

2.2.1.4 Temperature

T:20C This is the temperature (in °C) inside the platform, measured at the RelayBoard.

2.2.1.5 Uptime

00d:00h:02m:37s The bottom line shows the uptime since startup in days, hours, minutes and seconds.

2.2.2 The Info View

This view mode is not available in all robot types.



2.2.2.1 Version Information

The first line of this view shows the hardware version (HV) and the software version (SV) of the RelayBoard. For software updates please contact Neobotix.

2.2.2.2 Serial Number

The second line shows the serial number (SN) of the RelayBoard.

2.2.2.3 Status Information

The third line indicates the current configuration of the software and the state of the hardware components.

The asterisk (*) indicates that the RelayBoard did not yet receive a valid configuration from the control software. After configuration it will be replaced by a C.

The right part shows the state of all CAN bus devices. Motors are indicated by an M, the IOBoard by an I and the USBoard by a U. Devices that are not installed are marked by a dash (-) and devices that failed to connect are marked by an F.

2.3 Batteries

2.3.1 Battery Types

2.3.1.1 AGM Batteries

AGM batteries (Absorbent Glass Mat) are lead acid batteries that contain a fibreglass mesh between the battery plates. This mesh contains the electrolyte and prevents leakages even if the batteries are stored upside down or when the outer casing is cracked. This technology is well proven, cheap and widely available. Its drawbacks are a low energy density, resulting in high weight for a given capacity, and a limited number of charge-discharge cycles.

Their capacity will reduce over time until at some point (approximately after two years of use) the robot's available uptime is no longer sufficient and the batteries need to be replaced.

There are several measures to prolong the life of AGM batteries and to save costs and resources:

- Recharge the robot whenever possible.
- Avoid deep discharge and reaching the automatic shut-down voltage.
- Charge the robot while it does not need to move, for example while programming for several hours.
- Do not leave the robot with empty batteries for a long time.

Lead batteries in AGM design are generally very tolerant and significantly less dangerous than traditional lithium-ion-batteries. Heat, age or wrong treatment may nevertheless cause damages.

Attention: Regularly check the battery's condition, at least once every month! The following points indicate battery damages:

- Deformation or stretching of the battery's casing
- White or brownish spots at the battery contacts or other metal parts
- Excess heat
- Strong smell of sulphur
- Liquid below the battery

As soon as you notice any kind of damage the batteries must no longer be used and must not be charged any further! Please contact Neobotix immediately.

2.3.1.2 LiFePO4 Batteries

Lithium iron phosphate batteries, also known as LiFePO4 or LFP batteries, are a rather recent development in lithium-ion battery technology. They provide several benefits compared to AGM batteries as well as other lithium-ion designs.

LiFePO4 batteries are not necessarily smaller than AGM batteries but save about 60% of weight at the same nominal capacity. Under actual use conditions their usable capacity is even higher than that of AGM batteries, because they can be fully discharged. Similar deep discharge of AGM batteries quickly leads to loss of capacity or other damages.

Thanks to their chemistry LiFePO4 batteries provide a very stable output voltage and support fast charging. This not only reduces the recharge time significantly but also allows short periodic charging at convenient locations inside the workspace. When done properly this in-process-charging enables the robot to operate 24/7 without being taken out of operation for recharging.

LiFePO4 batteries are intrinsically much more stable than other lithium based designs. In addition our LiFePO4 batteries also contain a sophisticated battery management system (BMS) that constantly monitors their condition and health. The status information of the BMS is read by the robot's control software and used to indicate potential problems.

Warning: If you notice anything unusual about the LiFePO4 batteries immediately switch off the robot, stop charging and contact Neobotix!

2.3.2 Charging

2.3.2.1 With external Battery Charger

The battery of the mobile robot consists of several batteries connected in series and / or in parallel. The resulting battery pack can be charged conveniently and safely by using the provided battery charger.

The battery charger needs a power supply of 230 VAC, 50 Hz.

The charger must be placed in a dry environment. Do not cover the housing, so appropriate cooling is possible, and avoid direct sunlight.

First connect the battery charger to the charging connector and then turn on the charger. After the battery has been fully charged first turn off the battery charger and then disconnect the cable.

Warning:

- Never connect or disconnect the battery charger while it is switched on! Increased wear of the contacts due to electric arcs may occur. The battery or other electrical components may be damaged.
- Always make sure that the charging connector is fully inserted into the socket and properly secured. An incomplete or insufficient connection can cause high heat and significant damages.
- Only use batteries and chargers supplied by Neobotix. Using unsuitable batteries or battery chargers can lead to serious damages, injuries and hazards!

Recharging is done completely automatic after the connection is established and the battery charger is connected to the main power supply and switched on. For recharging it does not matter whether the robot is switched on or off. The only difference is the time needed for a full recharge. With the on-board electronic active, a recharge might take significantly longer, depending on the workload of the on-board computer.

The battery charger features an overload protection, thus allowing the robot to be permanently (e.g. overnight) connected to the charger. If the robot remains switched on while being connected to the charger (e. g. while programming) it might be necessary to switch the battery charger off and on once a day.

Please also read the documentation that was supplied together with the battery charger.

2.3.2.2 With integrated Battery Charger

Some robots feature an integrated battery charger. These robots can be charged by first plugging the charging cable into the robot's charging socket and then plugging the other end into a normal power outlet as described above.

Warning: Always make sure that the charging connector is fully inserted into the socket and properly secured. An incomplete or insufficient connection can cause high heat and significant damages.

The charging process starts automatically once the connection to mains has been made. As with the external charger the robot can be charged both when switched on or off.

2.3.3 Recycling

Both battery types used in Neobotix robots are leak proof and very safe.

Warning: They still contain highly reactive chemicals and must never be opened or damaged to avoid serious damage or injuries!

Attention:

- Under European law all kinds of batteries must only be returned to certified recycling companies. Please get in touch with Neobotix if you are unsure about how to recycle or dispose of used batteries.
- You can also return all old or wasted batteries from your Neobotix product to Neobotix free of charge for proper recycling.
- Please make sure to fully cover the battery contacts to guarantee their electrical insulation before shipping or disposing of old batteries. This will prevent short circuiting, damages and injuries.

2.4 Charging Stations

2.4.1 Automatic Charging Station

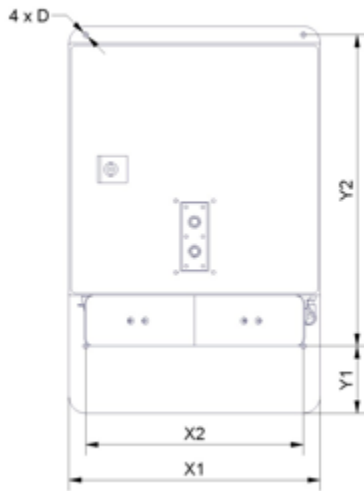
The automatic charging station is very robust and can easily be installed and integrated into the robot's program.

In order to allow a fast and trouble-free docking, there needs to be some free space to each side of the station and to the front. Find a place that can be easily accessed by the robot when it needs to recharge but in which neither station nor robot are obstacles to anybody. A power outlet must be nearby for the battery charger to be plugged in.

Please mind the following when choosing the stations mounting place:

- In order to have the charging contacts on the correct height, the bottom edge of the station's backplate must be placed directly on the floor.

- The charging station must be mounted to a stable wall. Please contact Neobotix if you need a free-standing charging station.
- The station must be mounted centred at the rear end of a free path that is at least 1.0 m wide.
- When using more than one station, please keep a grid of at least 1.0 m width.



2.4.1.1 Dimensions

The dimension of the charging station varies depending on the robot. The dimensions can be found in the table below:

Robot	X1	X2	Y1	Y2	D
MP-400	300	260	80	370	6.6
MP-500	300	260	110	360	6.6
MPO-700	300	260	192	370	6.6
MPO-500	340	320	40	290	6.6

2.4.1.2 After Installation

After installing the charging station please check the height of the charging contacts and if the robot can reach the station without problems.

Attention: Only plug in the power cable after successfully checking these two points. The charging station requires the same power supply as the external battery charger.

To allow automatic charging the battery charger inside the station needs to remain switched on at all times. Despite the open contacts which the robot needs for charging, there is no danger of electric shock or short-circuiting due to the charger's control mechanism. There will only be a current if the device has detected the correct batteries.

Note: In normal operation, the platform's charging contacts are disconnected from the batteries by a power relay on the platform's RelayBoard. This relay can be controlled by software.

Attention: The wear of the electrical components can be reduced by stopping the charging process only after the batteries have been fully charged and the charging current is reduced.

2.4.2 External Battery Charging Station

If the robot is equipped with the battery quick change system the battery set that is currently not in use can be recharged in an external battery charging station.

Place the charging station on the floor and make sure that it will not slip. The station can either be placed directly in front of a wall or can be screwed to the floor.

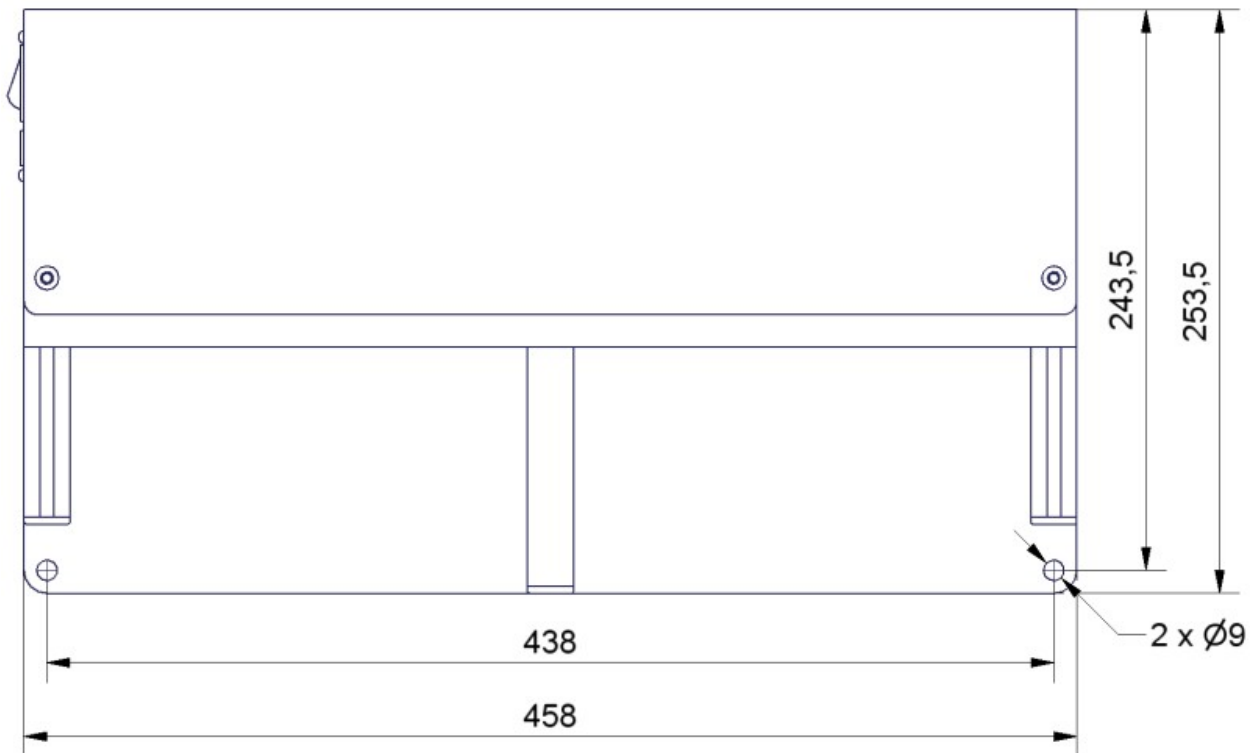


Fig. 1: Dimensions of the charging station

Make sure that the input voltage selection switch is set correctly and then connect the charging station to an ordinary power outlet. Before installing any batteries always make sure that the integrated battery charger is switched off. The main power switch is located directly beside the power cable connector.

Batteries can only be charged as a complete set of two identical batteries with the same charge level. Place the batteries onto the station's base plate between the aluminium profiles with the handles on top and the battery contacts facing towards the station's rear. Then slowly push the batteries into the station until they touch the vertical plate and are flush with the station's front.

The green LED at the side of the battery charger will light up constantly when the batteries are fully charged. They can remain inside the charging station until they are needed.

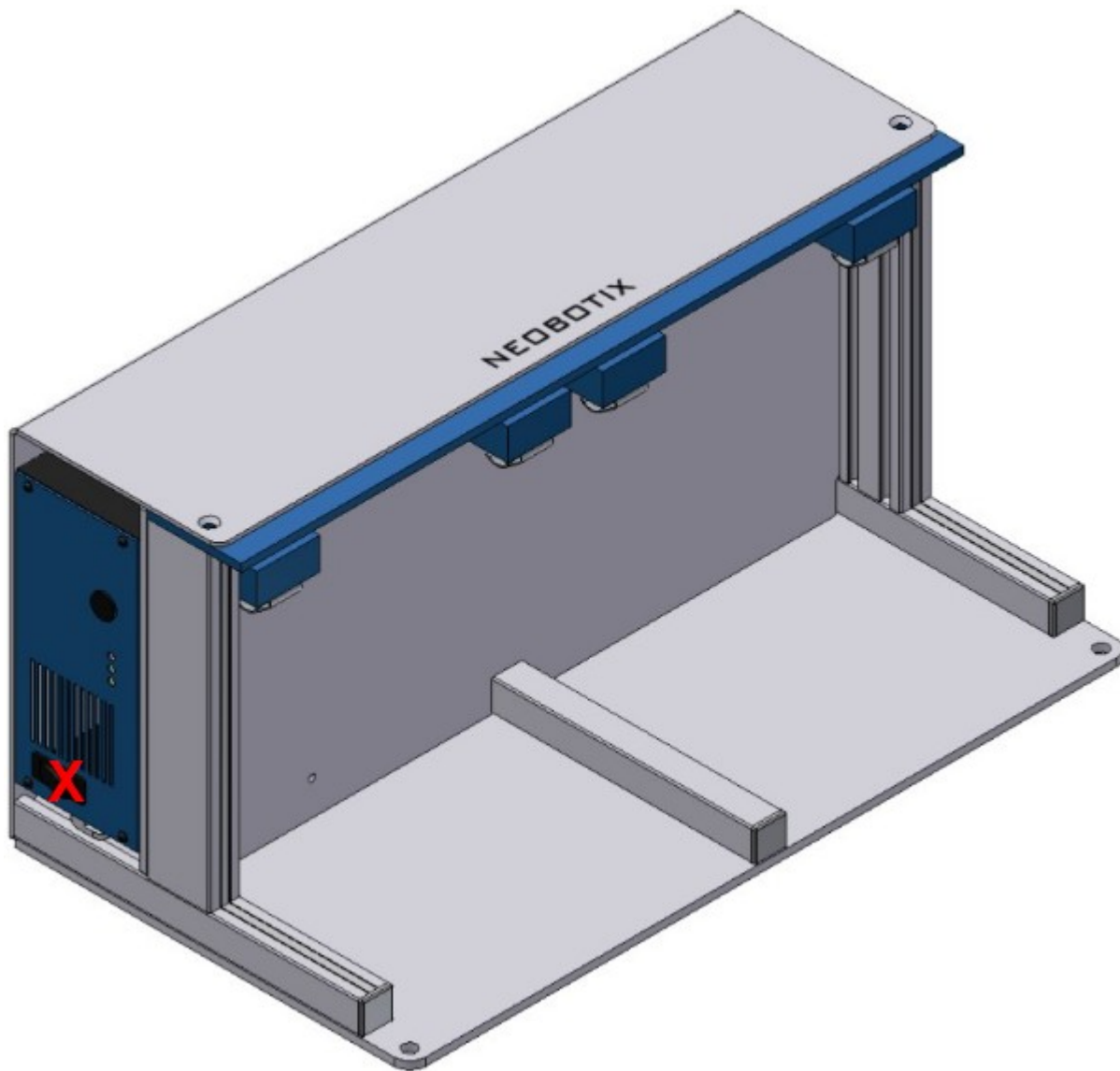
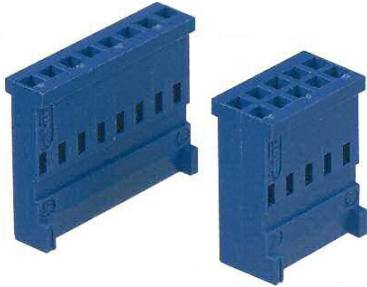


Fig. 2: Position of the main power switch (X)

2.5 Connectors

2.5.1 TE Connectivity - HE14

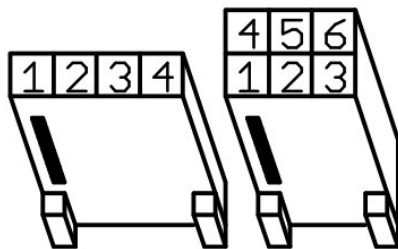


Pins	TE Connectivity	Farnell	RS Components
3 pins, 1 row	281838-3	429582	532-333
4 pins, 1 row	281838-4	429594	532-349
5 pins, 1 row	281838-5	429600	532-355
6 pins, 2 rows	281839-3	429650	532-406
8 pins, 2 rows	281839-4	429661	532-412
10 pins, 2 rows	281839-5	429673	532-428
12 pins, 2 rows	281839-6	429685	532-434



Crimp contacts	TE Connectivity	Farnell	RS Components
AWG 28-24	182734-2	429715	532-456

In Neobotix products the pin assignment of the HE14 connectors is as shown below.



2.5.2 Würth Elektronik - MPC4

Please check the [Würth Elektronik online catalogue](https://www.we-online.de/web/de/wuerth_elektronik/start.php)² for details on the MPC4³.

² https://www.we-online.de/web/de/wuerth_elektronik/start.php

³ https://www.we-online.de/katalog/de/em/connectors/wire-to-board/wr_mpc4/

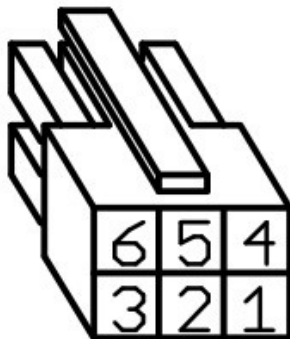


Pins (in 2 rows)	Würth Elektronik
2	649002113322
4	649004113322
6	649006113322
8	649008113322
10	649010113322
16	649016113322



Crimp contacts	Würth Elektronik
AWG 24-18	64900613722

In Neobotix products the pin assignment of the MPC4 connectors is as shown below.



2.5.3 Würth Elektronik - MPC3

Please check the [Würth Elektronik online catalogue](https://www.we-online.de/web/de/wuerth_elektronik/start.php)⁴ for details on the MPC3⁵.

⁴ https://www.we-online.de/web/de/wuerth_elektronik/start.php

⁵ https://www.we-online.de/katalog/de/em/connectors/wire-to-board/wr_mpc3/

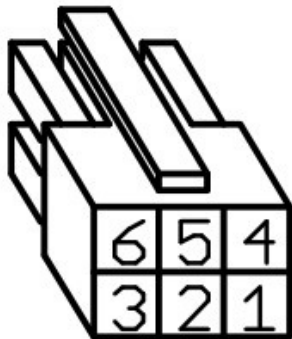


Pins (in 2 rows)	Würth Elektronik
4	662004113322
6	662006113322
12	662012113322



Crimp contacts	Würth Elektronik
AWG 24-20	66200113722

In Neobotix products the pin assignment of the MPC3 connectors is as shown below.



2.6 Safety Instructions

2.6.1 General Safety Instructions

This page contains general safety instructions and information that applies to all Neobotix robots.

Note: Depending on the chosen options and customisation of your robot additional safety instructions may apply. Please also check the documentation of the actual robot model that you are working on to ensure a safe and trouble free operation.

The mobile robot (in all varieties and models based on it) is an autonomous vehicle that is not only capable of performing very different tasks but can also react dynamically to its environment. Under some circumstances this might result in situations in which the robot's behaviour appears to be surprising or incomprehensible to persons who do not know the robot in detail.

Because of this it is essential that all of the following safety instructions are followed at all times. A safe and efficient operation of the robot can only be achieved under this condition.

2.6.1.1 Emissions

All the components and sensors used in Neobotix robots are safe to use and do not emit any dangerous radiation.

- The laser scanners are devices of laser safety class 1 or 1M.
- The ultrasonic sensors are safe for both humans and animals.
- None of the electronic devices and components will heat up to a dangerous degree.

2.6.2 Briefing

Due to the complexity of the robot system it is strongly recommended to attend a training. This online documentation is only provided as a work of reference when facing minor or common problems. All other problems, mistakes and difficulties can best be avoided by an in-depth training for programmers and a proper briefing for operators.

Attention: Always make yourself familiar with the robot, its control and behaviour before starting your work or setting up autonomous processes. Every other concerned person (e. g. workers, programmers, visitors etc.) should also be briefed accordingly.

2.6.3 Safety System

The mobile robot is equipped with several safety features. Additional features can be installed if necessary. Before starting to work with the robot a risk assessment may be required to determine whether the present safety features provide a sufficient level of safety.

Warning: Some of the safety features, especially the laser scanner, might have been reduced or even disabled in order to make transport to the customer's site possible or more convenient. These features must be set up and tested by a qualified technician prior to the first operation of the robot system!

2.6.3.1 Emergency Stop Buttons

As soon as one of the emergency stop buttons is pressed, all drives are immediately disconnected from power and the safety brakes of all motors (if equipped) are engaged.

Additional information on the emergency stop buttons and the reset procedure can be found in the specific documentation of your robot model.

2.6.3.2 Laser Scanners

The safety approved laser scanners can monitor user-defined safety fields in front of or around the robot which are dynamically activated according to the current velocity of the platform. If required these fields can also be activated by an application specific control software. In this case only the control software is responsible for the correct activation of the safety fields according to the current condition

Note: Please note that in omnidirectional robots the safety fields can only be activated by software and not depending on the robot's speed.

As soon as a person or obstacle is detected within the currently active safety field the robot is immediately set to emergency stop. The stop will be reset automatically after the field has been cleared. No manual reset is required in this case.

The laser scanners have been approved as safety device with Performance Level d and SIL2. They fully replace the mechanical safety bumpers which have been required for autonomous vehicles in their early days.

2.6.3.3 Special Functions

In case third-party software is meant to run on the robot or on the remote control computer this software can be enabled to trigger an emergency stop and or to reset the safety system. Additional safety devices can also be integrated into the mobile robot.

Please contact Neobotix if you plan to implement any special functions.

2.6.4 Cooperating with the Robot

2.6.4.1 Close Proximity

As long as the robot is not in full emergency stop, a minimum distance should always be kept. Direct contact to the active robot system is to be avoided.

Warning: Climbing up onto the mobile platform and / or riding on it are not allowed at any time.
--

2.6.4.2 Medium Proximity

When approaching the robot or working in its presence, increased attention and caution are required. The safety features of the mobile robot have to be set at such a level that imminent dangers can be detected and that injuries or damages can be avoided.

In case the safety features have to be reduced due to the requirements of the process it might be necessary to prohibit the presence of people in the robot's surrounding. In these cases only the owner of the robot system can be held responsible for the safe operation.

2.6.4.3 General Surrounding

As long as the safety features of the mobile robot are set up properly, the presence of people and vehicles in the robot's wider working area is allowed. Please contact Neobotix for further advice if needed.

All people working in the same area as the robot should nevertheless be informed about the robot's behaviour and the possible dangers. Appropriate behaviour and consideration are necessary.

2.6.5 Bringing into Service

After installing the mobile robot and also after making changes to the environment or the work processes a supervised test-run has to be performed. This test has to cover all possible steps of the process. Autonomous operation must not be resumed without a successful test-run.

This also applies to modifications of routes and target positions, parameters, environmental conditions and higher level control systems.

Repairs, maintenance work and other changes in the system's hardware should also be followed by a test-run.

Note: The Neobotix GmbH cannot be held responsible in any way for injuries or damages which are caused by any problem that could have been detected or prevented by a supervised test-run.

2.6.6 Modification of the System

If you intend to perform any kind of mechanical, electrical or software modification and feel unsure about the process, please contact Neobotix. Some detailed information or instruction might be necessary.

In case one or more robots are to be modified, it is strongly recommended to consult Neobotix in order to provide the appropriate training and information for all technicians and programmers. The functions and safety of all modified robots have to be checked and ensured before bringing them into service.

Attention: All warranties are void in case of any unauthorised or improper modification of the mobile robot system. All responsibilities for the further operation of the robot are devolved to whoever commands or executes such modifications.

2.6.7 Expected Misuse

2.6.7.1 Passenger Transportation

Danger: Transporting passengers on or by the mobile robot is dangerous and strictly prohibited. The Neobotix GmbH cannot be held responsible for any injuries and / or damages caused to or by transporting passengers with the mobile robot.

2.6.7.2 Safety Fields

The laser scanners' safety fields must be configured and tested by the operator prior to taking the robot into service. If more than one safety field are to be used they need to be dynamically activated according to the current state and intended action of the robot.

Warning: The operator takes full responsibility for the safe activation of the safety fields. Safety fields that are set up incorrectly or have been activated faultily will result in an unsafe and possibly dangerous operation of the mobile robot.

2.6.7.3 Working Area

When choosing and preparing the working area special care has to be taken to ensure both a reliable localisation and motion control. The robot must never be operated in areas where there are staircases leading down, elevated platforms or other possibilities for falling or dropping down.

The localisation system of the mobile robot requires clearly visible landmarks and a high quality map of the surrounding. Further information on this topic can be found in the documentation of the used software. Please contact Neobotix in case of any problems.

The motion control system of the mobile robot will only work reliably on an even ground and without slippage. Dirt like dust, sand, oil or water may cause the wheels to slip and the robot to move uncontrolled. This might even result in a loss of localisation which might make the robot leave the predefined path and enter prohibited or dangerous areas.

Danger: The robot must never be operated in areas where it might fall down staircases, loading ramps or similar.

2.6.7.4 Mechanical Overload

The defined maximum payload must not be exceeded. Overloading the robot may lead to reduced life expectancy of the drives and to damages to the robot. It might also affect the driving properties, resulting in unwanted movements and might even cause the robot to leave the predefined paths.

2.6.7.5 Electrical Overload

The robot's on-board power supply must not be overloaded. Extreme overload may lead to overheating, damages to the electrical installation and to short circuiting.

Please contact Neobotix before modifying the electrical system of the mobile robot.

2.6.7.6 Modifications

The robot must always be turned off and disconnected from all power sources before any modification is carried out. Working on the robot while it is connected to a power source or turned on results in the risk of electrical shock to the worker and damages to the robot.

Note: Additional information on maintenance and hardware modifications can be found at [Maintenance](#) (page 30).

2.6.8 Configuring the Safety Fields

To achieve both high agility and a safe operation, the safety fields of the laser scanners must be configured and activated according to the intended motion. Prior to the first operation the safety fields have to be defined with consideration of the application's requirements, the working environment and the desired speed. If the safety fields are to be actively selected instead of being automatically activated, the wiring may need to be changed and a routine to select and activate the safety fields has to be implemented in the control software.

Warning: At delivery a set of safety field is defined which will avoid collisions in most situations. These settings must be tested and, if necessary, be adjusted prior to the first operation. Otherwise the required level of safety cannot be guaranteed.

The configuration and diagnostics software CDS from Sick can be used to conveniently configure the scanners' safety fields. The configuration cable that was included in delivery can be used to connect the scanners to the serial port of any external computer running the CDS.

Note: The Sick default password is `SICKSAFE`.

2.7 Safety Instructions (Manipulators)

2.7.1 Robot Arms

2.7.1.1 Universal Robots

The robot arms from Universal Robots were designed for direct interaction with humans. Their safety features limit their speed and power and significantly reduce the potential to cause injuries or damages.

Warning: Nevertheless the robot arm may hit people or obstacles before its safety features come into effect. When equipped with a dangerous tool, serious injuries may be caused.

2.7.2 Laser Scanners

The robot arm may reach beyond the laser scanners' safety fields that were originally set up for transport and delivery. In order to ensure a safe operation of the mobile manipulator, the safety fields must be adjusted to cover the robot arm's work space whenever possible and to prevent collisions between the arm and any person nearby.

2.7.3 Working required from the Customer

Mobile manipulators are very complex robot systems. The applications in which they will be used could not be foreseen in detail during the development of the robot. Therefore, beside any modifications needed for certain research work or experiments, some very basic tasks must be completed before the mobile manipulator can be used safely.

- An assessment of risks must be performed by the customer to identify the different measures needed to guarantee a safe operation.
- All safety measures required for safe operation have to be implemented by the customer according to the assessment of risks and under the customer's responsibility.
- The higher level control software for the mobile manipulator has to be provided and installed by the customer. The safe operation of the robot greatly depends on the proper and reliable functioning of this software. Therefore it must be made sure that the robot's speed and workspace are limited to a safe level and that all axes are coordinated properly at all times.

In order to achieve a safe operation of the mobile manipulator, several measures might be required. The following list gives a short and incomplete overview of possible measures.

- Using the laser scanners area monitoring feature for collision avoidance.
- Adding safety bumpers to the robot.
- Limiting speed, power and work area of the robot arm.
- Adding covers and shielding on the mobile manipulator.

Note: Additional actions might be required to guarantee the appropriate level of safety!

2.8 Maintenance

2.8.1 Cleaning

Before any cleaning starts the robot should be set into emergency stop by pressing one of the emergency stop buttons. It can also be switched off altogether. This precaution is recommended both for cleaning the robot and for cleaning the robot's workspace.

Attention: If the robot is not set to emergency stop or switched off it might move in an unexpected way, causing collisions or injuries!

2.8.1.1 Wheels

Over time the wheels might pick up dirt which should be removed regularly. A thick layer on the wheels can cause vibrations and noise when the robot is moving.

A layer of dirt on the castor wheels or the trailing cable also may reduce the robot's capability to safely discharge electrostatic charge.

2.8.1.2 Sensors

Cleaning the various sensors is indispensable to assure that the robot can move safely and swiftly through its environment.

Attention: Be very careful not to scratch the cover of the laser scanner or the membranes of the ultrasonic sensors. Damaging these components might cause severe malfunctions!

2.8.2 Maintenance and Repairs

2.8.2.1 General Precautions

Please mind these general precautions that will make working on the robot safer and more efficient.

- Only qualified personnel may do maintenance, repairs or modifications on the robot. The requirements on qualified personnel can be found at *Qualified Personnel* (page 31).
- Before starting any work always unplug the mains connection and disconnect the robot from the battery charger and the automatic charging station.
- Disconnect all batteries from the robot.
- Follow the applicable common safety guidelines for working on electrical systems.

2.8.2.2 Wear and Tear

The mobile robot has been designed for longest possible operational life and is almost maintenance free. It is nevertheless advisable to regularly check the robot and all related devices and components for signs of wear.

- Mechanical damages to operating elements, covers and other components will reduce protection against contact. They might also cause delayed and surprising malfunctions.
- Damages, deformations or dirt on the drive and castor wheels can cause unsafe or inaccurate movements and lead to loss of localisation.
- Vibrations and increased noises during operation can be indicators of something chafing or of mechanical overload.
- Unusual heat can point to mechanical or electrical overload or to batteries that are worn out and should be replaced.
- A reduced uptime between charging cycles is often caused by old batteries that have lost a significant part of their capacity.

2.8.2.3 Repairs

In case of any questions regarding repairs or replacement parts please get in touch with Neobotix. We are always happy to support you.

Please mind the following points in any case:

- Only use replacement parts with features and properties identical to those of the original parts.
- Always mount all covers and hull parts properly and securely after finishing your work.
- Make sure to reconnect and fix all electrical connections, especially shieldings and earthing straps.

2.8.2.4 Batteries

The batteries need to be checked regularly and have to be replaced after some time. Additional information on the batteries can be found at [Batteries](#) (page 16).

2.9 Qualified Personnel

This product must only be modified, commissioned, operated and serviced by qualified personnel. Qualified personnel are defined as persons who

- due to their specialist training and experience have adequate knowledge for the work at hand,
- have been instructed by the responsible robot operator in the operation of the robot or its parts and the currently valid safety guidelines,
- are sufficiently familiar with the applicable official health and safety regulations, directives and generally recognised engineering practice (e.g. DIN standards, VDE stipulations, engineering regulations from other EC member states) that they can assess the work safety aspects of the product and
- have access to this manual and who have read it.

The following groups of persons are generally not considered qualified:

- Employees, interns or other academic staff not familiar with the product,
- visitors and guests,

- all members of other departments of the company or institution in which the product is operated.

This list is not intended to be exhaustive.

2.10 Taking out of Service

2.10.1 Disassembly

Once the mobile robot has reached the end of its lifetime it should be disassembled and its components should be recycled.

Tip: Before the robot can be disassembled the battery charger, any other external power supply and the batteries must be disconnected from the robot. It is strongly recommended to wait for at least 30 minutes after disconnecting all power sources before starting the disassembly. This will help to discharge any electrical energy that might be left in the system.

Disassembly of the robot should be done by qualified personnel only, preferably someone who has already worked on the robot in the past and knows its technical details. This will ensure

- fast and trouble-free disassembly,
- reduced risk of injuries or damages to components that are to be reused and
- proper sorting of parts according to materials and way of recycling.

More information on qualified personnel can be found at *Qualified Personnel* (page 31).

2.10.2 Recycling

2.10.2.1 Reusable components

Many components of the mobile robot (e. g. the servo motors and the amplifiers) have a very long life expectancy and will most probably still be usable when the overall system has reached the end of its lifetime.

Please check carefully which components can be reused immediately or in later projects and make sure that they are removed with care.

Tip: Reusing components does not only help the environment by reducing resource consumption but will also save a significant amount of money.

2.10.3 Mechanical parts

The frame and the other mechanical parts of the robot are mostly made from aluminium or steel. Both materials should be separated completely and can then be sold to certified recycling companies.

2.10.4 Electrical components

Electronic scrap and electrical wires are both a source of valuable resources as well as a threat to the environment and must not be treated like consumer waste.

Attention: All electrical wires and components must therefore be collected and sold to or deposited at the appropriate recycling facilities.

2.10.5 Batteries

Note: You can find information about the batteries and the recycling process at *Batteries* (page 16).

2.11 Legal Notes

2.11.1 Version Information

The English part of this online documentation has been translated and is not the original. Please refer to the German version in case of uncertainties or questions.

2.11.2 Liability

Every care has been taken in the preparation of this manual which represents the state of technology at the time of its composing. However, inaccuracies or omissions might occur. Please inform Neobotix in case you notice any.

The Neobotix GmbH cannot be held responsible for any technical or typographical errors and reserves the right to make changes to the product and manual without prior notice. Neobotix makes no warranty of any kind with regard to the material contained within this document, including, but not limited to, the implied warranties of merchantability and fitness for a particular purpose. Neobotix GmbH shall not be liable or responsible for incidental or consequential damages in connection with the improper use of one or more of the products described in this manual.

2.11.3 Downloads and Further Information

Additional information, data sheets and documentations, also for other products of Neobotix, can be found on our homepage <https://www.neobotix-robots.com/>.